

Database Development and Design

1.

a) 3 Ways of implementing the conceptual model

Option 1:

Creating one table for the superclass

This table would have:

- Its attributes and all the attributes from both the subclasses
- A subtype discriminator (empType) to distinguish between the two types of employees ✓
- In an instance where the Employee is a Doctor, the attributes of the Nurse will be null. ✓
- Attributes: empNo[pk], empFname, empLname, empDoB, empType, position, shiftHours ✓

Option 2:

Implementing all three tables and having subclasses associated with the super class through appropriate 1-1 relationships.

- Each subtype would have its unique attributes, as well as a foreign key from the parent class, which would be shown as [pk][fk] in the design. ✓
- Employee: empNo[pk], empFname, empLname, empDoB
- Doctor: empNo[pk][fk], position ✓
- Nurse: empNo[pk][fk], shiftHours

Subtype discriminator?

Option 3:

Implementing only the subtype classes

- Each subtype must have all the attributes of the supertype class and its unique attributes. ✓
- The primary key will remain the same as the one in the supertype class
- Doctor: empNo[pk], empFname, empLname, empDoB, position ✓
- Nurse: empNo[pk], empFname, empLname, empDoB, shiftHours. ✓

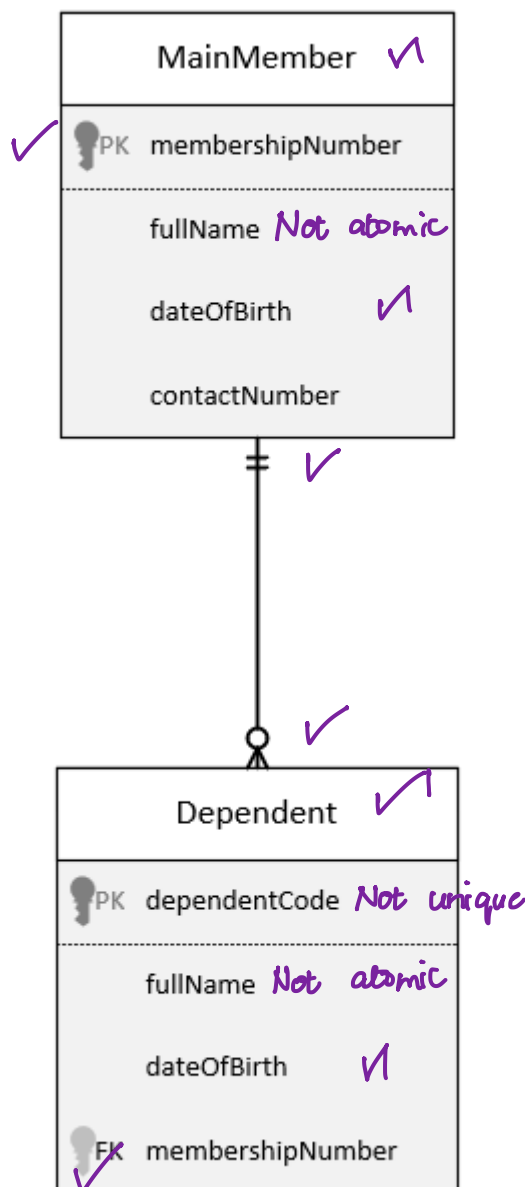
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- b) For the current situation, implementing all tables is best (Our option 2 above). Doctor and Nurse have at least one unique attribute, so to avoid nulls when **either(OR)** employee is instantiated, each class should have its own implementation. Both subclasses would now be associated with the parent class through a 1-1 relationship. This option will also reduce redundancy and update anomalies by storing shared employee data once in the Employee table and subtype tables storing only what is unique to them.

Is the supertype abstract in the scenario, or not?

2.

a)



b) Relational Schema

- MAINMEMBER (membershipNumber[pk], fullName, dateOfBirth, contactNumber) ✓✓
- DEPENDENT (dependentCode[pk], fullName, dateOfBirth, membershipNumber[fk]) ✓✓

Only fk?

Not atomic

3½